

Department of Computer Science



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Detection and Classification of Breast Cancer in Ultrasound and Mammogram Images

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Executive Summary

Aims and Motivation. Breast cancer is currently the most frequently diagnosed malignancy among women worldwide, accounting for approximately 31% of all new female cancers[1]. While mammography is the "gold standard" for early detection, its efficacy is significantly reduced in dense breast tissue; conversely, ultrasound (USG) is highly accessible but suffers from high operator-dependency. This project is motivated by the critical need to automate the detection and classification of lesions to provide a reliable "second opinion". The primary aim is to bridge the research gap between traditional Convolutional Neural Networks (CNNs) and emerging Vision Transformers (ViTs) by evaluating their performance under unified experimental conditions across both mammography and ultrasound modalities.

Methodology and Approach. The project utilized a structured Machine Learning pipeline, bifurcated into two workstreams: three-class classification (Benign, Malignant, Normal) and USG lesion segmentation. The approach was grounded in clinical integrity, employing:

Rigorous Preprocessing: Implementation of the Telea algorithm for automated inpainting to remove technician annotations and ROI cropping to eliminate non-diagnostic background noise.

Diverse Architectures: A comparative evaluation of state-of-the-art models, including ConvNeXt-Small (CNN), Swin-Transformer (ViT), EfficientNet-V2, and MaxViT for classification and combinations of ResNet and MiT-B3 with Unet/Unet++ for segmentation.

Advanced Training Protocols: A strategy involving a locked-backbone warm-up followed by fine-tuning with differential learning rates and Stochastic Weight Averaging (SWA) to enhance generalization.

Hardware Optimization: Deployment of Gradient Accumulation and Automatic Mixed Precision (AMP) to achieve SOTA performance on consumer-grade hardware.

Key Results. When evaluated on the standard BUSI dataset, the USG segmentation pipeline achieved a reproducible average Dice score of 0.90 (Precision: 0.92). The USG classification framework reached an F1-score of 0.95 and an AUC of 0.9890 on the same benchmark, confirming that the developed methodology is competitive with SOTA performance and is achievable using Transfer Learning on consumer hardware. In terms of mammography classification, the framework achieved an AUC of 0.68 on

the INBreast dataset, offering the most significant avenue for improvement.

Evaluation Against Success Criteria. The project successfully met its success criteria through 5-fold stratified cross-validation, ensuring statistical significance for all reported metrics. Performance was rigorously measured against the primary success metrics: Macro-Averaged F1-score and AUC-ROC for the three-class classification task, and the Dice Similarity Coefficient (DSC) and 95th Percentile Hausdorff Distance (HD95) for USG lesion segmentation.

- **Classification Frontiers:** ConvNeXt-Small proved to be a robust "High-Frequency" detector, benefiting from ROI cropping (lifting USG F1 to 0.8035) and SWA. However, Swin-Transformers acted as "Structural" detectors, providing better global context but showing performance degradation under SWA.
- **Segmentation Excellence:** The MiT-B3 encoder coupled with Focal Loss emerged as the performance leader, achieving a peak Dice score of 0.7733. Large effect sizes (Hedge's $g > 1.0$) confirmed that Transformer-based encoders benefit from stable, uniform learning rate schedules.

Clinical Implications and the Modality Gap. The identification of a persistent "Modality Gap", where ultrasound (F1: 0.7909) consistently outperformed mammography (F1: 0.7149), carries substantial implications for clinical deployment. This gap, driven by mammography's 2D projection "tissue masking" and ultrasound's superior contrast gradients, suggests that current single-modality CAD systems are more robust when applied to USG. For clinical practice, this means:

Trustworthiness through Interpretability: The research demonstrates that CNNs act as "High-Frequency" detectors (localized activations), which may be more "trustworthy" for radiologists than the broader structural context captured by Transformers. Grad-CAM analysis ensures decisions are based on pathological features rather than non-diagnostic artifacts.

Accessibility and Equity: By optimizing for consumer-grade hardware through Gradient Accumulation and Mixed Precision, these diagnostic frameworks are technically viable for deployment in low-resource clinical settings, lowering the barrier to entry for advanced "second opinion" tools.

Future work will focus on developing architectures and integrating them into end-to-end CAD systems to close the "Modality Gap".

Statement of Ethics

Avoidance of Harm and Clinical Trustworthiness

The fundamental principle of this project is the avoidance of harm, focusing on the welfare and interests of the wider community. By developing a Computer-Aided Diagnosis (CAD) system as a "second opinion" tool, the research aims to mitigate risks of misdiagnosis caused by ultrasound operator-dependency and the reduced efficacy of mammography in dense tissue. To prevent clinical complacency or "ill" application, the study emphasizes clinical trustworthiness through Grad-CAM interpretability analysis, ensuring model decisions are based on pathological features rather than non-diagnostic artifacts.

Data Protection and Informed Consent

In accordance with Articles 2.3 and 2.5 of the University Code of Ethics, this research utilized secondary, pre-anonymised medical datasets following formal approval from the Departmental Ethics Committee. All data handling complied with the Data Protection Act (2018); patient IDs were used strictly for stratified splitting to ensure data integrity without revealing individual identities. The implementation of automated inpainting to remove technician annotations further de-identifies the images while preventing the model from "cheating" through shortcut learning. This use of established research repositories justifies the deviation from direct written consent as allowed by university policy.

Social, Professional, and Commercial Implications

The project addresses a significant social challenge, as breast cancer accounts for 31% of all new female cancers. By optimizing models for consumer-grade hardware through Gradient Accumulation and Mixed Precision, the study promotes accessible and cost-effective diagnostic frameworks. Professionally, the research maintains academic integrity by employing a rigorous 5-fold cross-validation methodology to defend against data leakage. Commercially, the study provides a "safety warning" for industry adoption by identifying the "brittleness" of specific architectures like Transformers when subjected to certain optimization techniques, ensuring that future commercial tools prioritising global context do not sacrifice diagnostic precision. No significant environmental or cultural heritage issues were identified within the scope of this computational study.

1 Introduction

1.1 Motivation

Breast cancer remains a pervasive global health challenge and is currently the most frequently diagnosed malignancy among women worldwide. According to recent epidemiological data, it accounts for approximately 31% of all new female cancers, with incidence rates rising by roughly 1% annually, largely driven by increases in localized-stage diagnoses [1]. While multiple imaging techniques exist for aiding in early diagnosis, mammography and ultrasound serve as the primary screening and diagnostic modalities [2] [3].

Mammography is currently recognized as the "gold standard" for early detection, particularly for Stage 0 breast cancer, due to its high sensitivity to microcalcifications [2]. However, its efficacy is significantly reduced in women with dense breast tissue, a limitation often addressed by supplementary ultrasound screening [4]. Ultrasound uses high-frequency sound waves to visualize tissue structures and is highly accessible, non-invasive, and radiation-free. Despite these advantages, ultrasound is intrinsically operator-dependent; image quality varies significantly based on the technician's skill, the probe pressure applied, and the specific device settings used [3].

To mitigate these variabilities, Computer-Aided Diagnosis (CAD) systems have been developed to assist radiologists by providing a "second opinion." In recent years, the backbone of CAD has shifted from handcrafted feature extraction to Deep Learning (DL), which can automatically learn complex representations from raw data [5]. However, real-world deployment is hindered by complex image patterns, variable lesion sizes, and the "fuzzy" boundaries caused by acoustic shadows and speckle noise.

1.2 Aims and Objectives

Addressing the above points, the goal of this investigation is to develop an accurate CAD system capable of finding and classifying breast lesions in mammogram and ultrasound images. Different approaches will be compared of detecting and classifying breast cancer in the two modalities. The quality and quantity of the data required for training will be discussed alongside a comprehensive ablation study to arrive at empirically 'best' models for the task.

To comprehensively approach this aim, this investigation is split into 2

work streams, classification and segmentation. Separate experimental conditions and methodologies will be designed for these two different tasks to provide due considerations of their differing requirements. The classification task is to assign a numerical class to a given image of a certain modality from 3 options: Benign, Malignant, or Normal. On the other hand side, segmentation is concerned with detecting and outlining an area of the input image which corresponds to the lesion.

The investigation will focus on enhancing knowledge about training different types of architectures and cutting edge methodologies to maximize their performance in difficult scenarios.

1.3 Structure of Report

Appropriate methods and tools based on a robust Literature Review, Chapter 2, will be analyzed to establish a sound Methodology in Chapter 3 along with a comprehensive set of Research Questions and Success Criteria used to evaluate the results. A highly modular and reproducible Implementation of the Methodology will be discussed in Chapter 4, including common challenges and limitations. The achieved Results and Evaluation will be presented and discussed in depth in Chapter 5 alongside an analysis of obtained performance against the State of the Art. Finally, the Conclusion, Chapter 6, will evaluate the results of this study in respect to the above stated goal and present possible avenues for future work.

2 Literature Review

This Literature Review explores the evolution and current State of the Art (SOTA) methodologies used in Computer-Aided Diagnosis (CAD) systems. It examines the evolution of architectures and paradigm-shifting changes that have made Deep Learning (DL) instrumental to the widespread deployment of CAD systems thanks to increasing accessibility and accuracy. The review highlights how these systems enable more accessible preventative healthcare worldwide.

2.1 Evolution of Deep Learning in Breast CAD

Medical Image Segmentation is the foundational step in many CAD pipelines. It involves partitioning a digital image into multiple segments (sets of pixels) to precisely locate and delineate the Region of Interest (ROI) - in this case, the breast lesion. This technique provides critical geometric and volumetric information that is difficult to quantify manually due to the low contrast of ultrasound images. Deep learning models, predominantly variants of the

2 Literature Review

U-Net architecture, have become the SOTA for this task [5] [6].

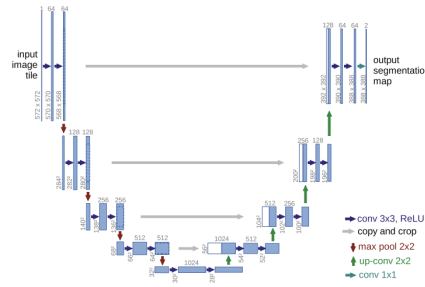


Figure 2.1: Sample 32x32 Unet architecture[6]

As illustrated in Figure 2.1, these models utilize an encoder-decoder structure with skip connections to preserve spatial information, learning to assign a class label (e.g., lesion or background) to every single pixel. Accurate segmentation is critical for subsequent processing, as it isolates the lesion for robust morphological feature analysis (e.g., assessing boundary irregularity or orientation), which directly correlates with malignancy [7]. The performance of these segmentation models is typically evaluated using overlap-based metrics such as the Dice Similarity Coefficient (DSC) and the Jaccard Index, which quantify the agreement between the predicted mask and the clinician's ground truth annotation [5] [8].

Following segmentation, or sometimes applied directly to the original image patch, Image Classification aims to assign a categorical diagnostic label to the detected region. For breast ultrasound, the primary objective is to classify the lesion according to the BI-RADS (Breast Imaging-Reporting and Data System) standardized categories (e.g., Benign, Malignant, or a specific score from 1 to 6) [7].

Modern approaches rely heavily on Convolutional Neural Networks (CNNs), such as ResNet or Inception architectures. These networks automatically learn a hierarchy of features, progressing from simple edges and textures in early layers to complex, abstract representations in deeper layers, that are predictive of the final diagnosis [9] [10]. The output is typically a probability score for each class, which supports the clinician's decision-making process. While these models often achieve high Area Under the ROC Curve (AUC) scores on controlled datasets, a significant challenge remains in ensuring that the learned features are clinically relevant and robust across different patient populations and imaging protocols [11].

Residual Learning and Efficient Architectures

ResNet (Residual Networks) revolutionized this landscape by introducing "skip connections," which allow gradients to flow through the network without attenuation by bypass-adding the input of a block to its output[9].

In the context of breast imaging, AnbuDevi (2022)[12] utilized ResNet-50 on the BUSI dataset[13], demonstrating that residual learning enabled the model to converge faster on subtle morphological features of malignant tumors than traditional VGG architectures. Building upon this, EfficientNet introduced "Compound Scaling," which scales width, depth, and resolution simultaneously using a fixed ratio to optimize parameter efficiency. Khan et al. (2020)[14] demonstrated that an EfficientNet-B2 could outperform heavier models like ResNet-152 on mammogram classification while utilizing 80% less memory, a critical factor for processing high-resolution medical images where fine details like microcalcifications are diagnostic.

Segmentation and Spatial Preservation

For delineation tasks, U-Net remains the benchmark due to its symmetrical encoder-decoder structure. Its "concatenation skip connections" preserve high-resolution spatial features from the initial layers, which is vital for delineating the often-ambiguous boundaries of USG lesions. Litjens et al. (2017)[15] noted that while the encoder path captures high-level semantic context ("what"), the decoder path uses skip connections to recover precise spatial localization ("where").

2.2 Emerging Paradigms

While CNNs have dominated the field, they are fundamentally limited by their "local receptive field," which prioritizes local textures over global structural context. This has led to the emergence of Vision Transformers (ViTs) in the medical domain. [16]

Global Context vs. Local Texture

Unlike CNNs, ViTs treat an image as a sequence of patches and utilize self-attention mechanisms to capture global dependencies across the entire frame. Gheflati and Rivaz (2022)[17] conducted a comparative study of ViTs in breast ultrasound, finding that while ViTs showed superior robustness to speckle noise, their accuracy was highly sensitive to the quality of pre-training.

Furthermore, Dosovitskiy et al. (2021)[18] established that ViTs lack the "inductive biases" inherent to CNNs, such as translation invariance, making them "data-hungry." To mitigate this, Meng et al. (2024)[19] proposed hybrid architectures like TransUNet, which utilize CNNs for low-level feature extraction and Transformers for global relationship modeling. This hybrid approach achieved a Dice Similarity Coefficient (DSC) of 0.82 on USG datasets, outperforming standard U-Nets in cases where tumors exhibited complex posterior shadowing.

The Role of Transfer Learning (TL)

Given the scarcity of expert-annotated medical data, Transfer Learning is ubiquitous. By pre-training on massive datasets like ImageNet, models learn fundamental visual features (edges, textures, and shapes) that are transferable to medical imaging [20], [21]. Byra (2019)[22] reported that ResNet-50, when fine-tuned via TL, achieved diagnostic accuracy exceeding 90% on controlled breast lesion benchmarks, significantly outperforming models trained from random initialization.

However, the application of ImageNet weights to medical imaging is not without domain-shift challenges. Significant discrepancies exist between the 3-channel RGB source domain and the 1-channel (greyscale) target domain typical of medical modalities. Research by Raghu et al. (2019) [23] suggests that while ImageNet pre-training speeds up convergence, its performance benefits are often marginal for medical tasks because the high-level features learned from natural objects do not always represent anatomical structures effectively. To mitigate this feature-level shift, common practices include replicating the greyscale channel across RGB inputs or employing "domain-specific" pre-training on larger unlabelled medical datasets to better align the model's internal representations with clinical data distributions [24].

2.3 Critical Challenges

Despite architectural advancements, significant gaps remain in clinical deployment, particularly regarding image quality and population-level bias.

Modality-Specific Artifacts

Ultrasound imaging is uniquely susceptible to inter-operator variability and intrinsic degradation via speckle noise, acoustic shadowing, and enhancement [25]. Geirhos et al. (2022)[26] argued that CNNs are prone to "texture bias," where the model overfits to these local noise patterns rather than the underlying pathology. This underscores the theoretical advantage of ViTs, which may offer greater robustness to local corruptions through global shape modeling [27].

Class Imbalance and Population Heterogeneity

A pervasive issue in medical datasets is the disparity between benign and malignant samples (often an 80/20 split). Sun et al. (2025)[28] observed that models trained on imbalanced data often yield high overall accuracy but dangerously low sensitivity (recall) for malignant cases, essentially learning to predict the majority class.

Furthermore, Rabiei et al. (2022)[29] highlighted a systematic research gap regarding breast density and demographic diversity. Models trained

primarily on low-density tissue (BI-RADS A and B) demonstrate significant performance degradation when tested on dense breast tissue (BI-RADS C and D), which is more prevalent in certain ethnic and age groups.

Scarcity of Multi-Modal Frameworks

While the clinical synergy between Mammography (MG) and Ultrasound (US) is well-established, deep learning research remains largely siloed by modality-specific benchmarks. In mammography, the CBIS-DDSM [30] serves as the primary large-scale repository for mass and calcification detection, while the INbreast database [31] is favored for high-fidelity digital evaluation. Conversely, ultrasound research is dominated by the BUSI dataset [13].

A novel field aiming to provide an end-to-end approach to breast cancer diagnosis is the use of Multimodal Large Language Models (MLLMs). These highly complex and extremely large models process and analyze various forms of data, including different modalities of breast cancer screenings, to enhance diagnostic potential [32]. Rao et al. (2023) [33] explored the use of OpenAI's GPT-4 as an MLLM, achieving a 98.4% on breast cancer screening prompts by using multi-modal data and unstructured data such as pathology reports and clinical notes obtained from electronic health records (EHRs). On the other hand, Guo et al. (2024) [34] proposed KAMnet, a multimodal classification framework trained on two types of ultrasound images (contrast-enhanced and B-mode) to achieve an accuracy of 88.24% on a custom dataset.

These achievements highlight how the SOTA is slowly moving towards unified frameworks which can process various forms of data to seamlessly integrate with clinical professionals.

2.4 Chapter Summary

The progression of breast CAD has been defined by a transition from handcrafted features to DL architectures that automatically extract complex pathological representations. While Convolutional Neural Networks (CNNs) like ResNet and EfficientNet have established a robust foundation for local texture analysis and parameter efficiency, they are fundamentally constrained by a "local receptive field" that may overlook broader anatomical relationships. Conversely, Vision Transformers (ViTs) offer a promising paradigm shift toward capturing global structural context, yet they remain "data-hungry" and sensitive to the specificities of medical data distributions.

Despite these advancements, the literature reveals several critical bottlenecks in clinical translation:

Modality-Specific Vulnerabilities: Ultrasound remains hindered by operator-dependent noise and acoustic artifacts, while mammography efficacy is compromised by glandular tissue density.

Optimization and Bias: There is limited understanding of how differing inductive biases, local vs. global, react to standard clinical preprocessing (such as ROI cropping) or advanced optimization techniques like Stochastic Weight Averaging (SWA).

Evaluation Disparity: Most existing studies evaluate architectures in isolation or on single modalities, often failing to account for the class imbalance (Benign vs. Malignant) pervasive in real-world clinical datasets.

Further investigating the behavior of different SOTA models (both CNNs and ViTs) on various modalities under uniform experimental conditions can inform critical insights in moving towards unified frameworks touched upon above.

3 Methodology

This chapter delineates the systematic approach employed to compare breast cancer classification performance across Ultrasound (USG) and Mammogram modalities, alongside an investigation into USG image segmentation to create a reproducible pipeline for training CAD systems. The project follows a structured Machine Learning Pipeline, an adaptation of the Agile method, encompassing data acquisition, rigorous preprocessing, and iterative experimentation and continuous reanalysis.

3.1 Research Gap

Based on the SOTA explored in the Literature Review, Chapter 2, Research Questions have been formed to guide this exploration:

Modality Performance Variance: To what extent does the diagnostic accuracy of state-of-the-art deep learning models vary between mammography and ultrasound (USG) when evaluated under unified experimental conditions?

Clinical Preprocessing Utility: What is the marginal utility of automated ROI cropping and artifact inpainting in stabilizing model focus and preventing "shortcut learning" from non-diagnostic technician annotations?

Architectural Inductive Biases: How do the differing inductive biases of modern CNNs (e.g., ConvNeXt) and Vision Transformers (e.g., Swin) impact their diagnostic interpretability and their sensitivity to optimization techniques like Stochastic Weight Averaging (SWA)?

Segmentation Frontiers: Which combination of encoder architectures (CNN vs. Transformer), loss functions, and optimization strategies yields the most robust performance for high-fidelity spatial localization in breast ultrasound lesions?

The investigation is bifurcated into two primary workstreams:

Classification: A comparative analysis using an ablation framework to evaluate four state-of-the-art architectures on both USG and Mammogram datasets.

Segmentation: An exploration of USG lesion delineation using varying encoders, loss functions, optimization strategies, image sizes and pre-processing approaches.

3.2 Data Sources

The integrity of a medical imaging model is predicated on the quality and handling of its data, necessitating a selection process that aligns with clinical realism and architectural benchmarking. The following datasets were selected to facilitate a multi-modal comparative analysis between the clinical "gold standard" of mammography and the highly accessible, yet operator-dependent, ultrasound (USG) modality. This diversity is essential for evaluating whether emerging paradigms like Vision Transformers (ViTs) can maintain performance across different high-noise medical environments.

The composition of the selected data, comprising the BrCaWisconsin, BrEaST-Lesions-USG, and Breast Ultrasound Dataset for Cancer Diagnoses (BUDaCaD) repositories, was strategically chosen to satisfy both the classification and segmentation workstreams. These specific sources provide the high-fidelity ground truth masks required for lesion delineation and a sufficient volume of samples (e.g., 2,537 USG classification images) to mitigate the "data-hungry" nature of Transformer-based architectures. From an ethical standpoint, the use of these clinically validated repositories ensures that the research respects patient privacy and autonomy while benefiting from the "collective wisdom" of previously consented clinical populations.

To ensure a systematic analysis, the datasets were pooled and filtered based on the following justifications:

Clinical Representation: The inclusion of Benign, Malignant, and Normal categories across all modalities allows for a robust evaluation of model sensitivity and the impact of class imbalance.

Architectural Benchmarking: Utilizing multi-source data provides a rigorous test for the inductive biases of CNNs and ViTs, determining if they can generalise across varying patient populations and imaging protocols.

Technical Robustness: The availability of patient-level metadata across these sources is critical for preventing data leakage, as it allows for partitioning the 5-fold cross-validation strictly by patient identity rather than image count.

Finally, 3 pooled training sets were utilized and are available in A.1:

Pooled Mammography Classification: 536 from BUDaCaD (DCL) and 1061 from BUDaCaD (Spectra). 577 Benign, 473 Malignant, 537 Normal.

Pooled Ultrasound Classification: 760 images from BrCaWisconsin and 1777 from BUDaCaD (DCL). 1199 Benign, 314 Malignant, 1024 Normal.

Pooled Ultrasound Segmentation: 760 images from BrCaWisconsin and 252 from BrEaST-Lesions-USG. 421 Benign, 208 Malignant, 131 Normal (not all images in this dataset have a class label).

3.2.1 Pre-processing and Ethical Data Integrity

To ensure model robustness and prevent the deployment of clinically misleading systems, a rigorous pre-processing pipeline was developed. Each step serves a specific ethical and technical purpose:

Inpainting (Technician Artifact Removal): Ethically, this prevents the model from "shortcut learning." If a model learns to identify malignancy based on a technician's digital highlight or annotation rather than the physiological features of the lesion, it becomes clinically dangerous. Inpainting ensures the model's decisions are based on pathological evidence, upholding the principle of non-maleficence by preventing false confidence in a biased system.

Patient-Level Splitting: This is a core requirement for scientific integrity. Partitioning by patient identity rather than image count prevents data leakage, where a model "memorizes" a specific patient's anatomy across training and validation sets. Ethically, failing to do this results in "optimism bias," where reported performance is artificially inflated, potentially misleading clinicians about the model's true diagnostic reliability.

De-duplication (DHash): Ensuring no image overlap between sets is an ethical safeguard against data contamination. It ensures that the evaluation of the model is a true measure of its ability to generalize to "unseen" patients, which is the primary ethical requirement for any tool intended for clinical decision support.

CLAHE and ROI Cropping: By enhancing lesion boundaries and focusing on pathological features, these steps improve the interpretability of the model. In a medical context, a model that focuses on irrelevant background noise is less transparent, making it harder for a clinician to trust or verify the output.

3.3 Experimental Design

This methodology utilizes three systematic ablation studies to quantify the marginal utility of architectural backbones, loss functions, and processing techniques. By varying these components, the study isolates the variables most critical to diagnostic performance in complex medical environments.

3.3.1 Classification Ablations

Four architectures, **ConvNeXt-Small**[35], **Swin-Transformer**[16], **EfficientNet-V2-S**[36], and **MaxViT-Tiny**[37], were selected to evaluate distinct inductive biases (convolutions, hierarchical attention, and hybrids) under medical data constraints.

To ensure clinical and ethical validity, "Small" and "Tiny" variants were utilized to maintain a robust parameter-to-sample ratio and prevent overfitting. This choice also promotes clinical equity, ensuring SOTA performance is achievable on standard hardware in low-resource settings rather than requiring industrial GPU clusters. These models were further chosen for their ability to handle multi-scale pathology, from microcalcifications (Swin/MaxViT) to large masses (ConvNeXt).

Experiments followed a two-phase approach (Table 3.1): first, isolating data and loss modifications on a baseline, then evaluating architectural performance. To address class imbalance, Focal Loss (α -weighted) was implemented to prioritize "hard" malignant examples. Finally, Stochastic Weight Averaging (SWA) was applied to locate flatter regions of the loss landscape, enhancing generalization against the distributional shifts common in clinical test data.

Table 3.1: Systematic Ablation Design for Classification

Experiment ID	Component Changed	Rationale
No Augmentation Baseline	ConvNeXt-Small + CE Loss	Establish performance on raw data
Baseline	ConvNeXt-Small + CE Loss	Establish augmented performance floor
Ablation 1	+ ROI Cropping	Evaluate impact of background noise removal
Ablation 2	+ Focal Loss (α weights)	Address significant class imbalance
Ablation 3	Switch to EfficientNet-V2-S	Evaluate parameter-efficient scaling
Ablation 4	Switch to MaxViT-Tiny	Evaluate local/global hybrid attention
Ablation 5	Switch to Swin-Transformer	Evaluate hierarchical windowed attention
Final Model	ConvNeXt + ROI + Focal + SWA	Maximize generalization via weight averaging

3.3.2 USG Segmentation Ablations

Segmentation ablations (Table 3.2) contrast a traditional CNN (ResNet34) with a modern Mix Transformer (MiT-B3) to assess self-attention’s impact on boundary precision. The study evaluates the synergy between BCE (pixel-wise accuracy) and Dice/Tversky losses; the latter provides the ethical safeguard of penalizing false negatives to avoid under-segmenting malignant lesions.

Input resolution was standardized to 512×512 to accommodate VRAM constraints during transformer training. While this "precision ceiling" may marginally smooth fine-grained boundaries (e.g., malignant spiculation), it represents a pragmatic ethical trade-off to ensure model accessibility on standard clinical hardware while maintaining sufficient fidelity for region-based optimization.

Table 3.2: Ablation Design for USG Segmentation

Experiment Groups	Encoder/Decoder	Key Variables Tested
Baselines	ResNet34 / MiT-B3	AdamW vs. AdamW Simple
Loss Ablations	UNet / UNet++	BCE vs. Dice vs. Tversky vs. Focal
Sensitivity	224 vs. 512 vs. 1024	Resolution, Preprocessing Agression, LR

These variations (available in full in Appendix A.1) allow for a granular analysis of how the objective function handles class imbalance (tumor vs. background) relative to the architectural "engine."

3.4 Success Criteria

To provide a statistically significant comparison, models are evaluated using 5-Fold Stratified Cross-Validation. For classification, this approach ensures that each fold maintains the original class distribution, mitigating the impact of class imbalance, which was particularly high in the USG Classification dataset. In the case of segmentation, the Folds are calculated to balance mask area size, ensuring each fold holds a comparable number of similarly sized masks to ensure accurate results.

3.4.1 Classification Metrics

The classification performance for the three-class task (Benign, Malignant, and Normal) is evaluated using metrics that account for both label accuracy and class imbalance. Model outputs are first transformed into class probabilities using a Softmax activation, while discrete class assignments are determined via the *argmax* operator. The primary performance metric is the Macro-Averaged F1-score. The F1-score is defined as the harmonic mean of precision and recall [5]:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (3.1)$$

To account for potential class imbalances across the diagnostic categories, Macro-averaging is utilized. This calculates the F1-score independently for each of the K classes and subsequently computes their arithmetic mean:

$$F1_{macro} = \frac{1}{K} \sum_{i=1}^K F1_i \quad (3.2)$$

This approach ensures that each diagnostic category contributes equally to the final score, regardless of its frequency in the dataset.

To assess the diagnostic discriminative ability of the classifiers across varying thresholds, we report the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Given the multi-class nature of the dataset, a One-vs-Rest (OvR) strategy is employed. In this configuration, the AUC is calculated for each class against all others, and the results are macro-averaged [5]:

$$AUC_{total} = \frac{1}{K} \sum_{i=1}^K AUC(C_i \text{ vs } C_{rest}) \quad (3.3)$$

For each ablation, the mean and standard deviation (σ) are reported across all cross-validation folds to ensure the robustness of the findings. Furthermore, the average validation loss is calculated by normalizing the cumulative loss by the total number of samples N in the dataset:

$$\text{Avg. Loss} = \frac{1}{N} \sum_{j=1}^N \mathcal{L}(\hat{y}_j, y_j) \quad (3.4)$$

3.4.2 Segmentation Metrics

To rigorously evaluate the segmentation performance and the impact of different ablations, a suite of overlap, boundary-based, and efficiency metrics are employed. For all pixel-wise calculations, a small smoothing constant

($\epsilon = 10^{-7}$) is added to the numerators and denominators to ensure numerical stability and prevent division by zero. The primary measure of overlap is the Dice Similarity Coefficient (DSC), which calculates the spatial overlap between the predicted mask (X) and the ground truth (Y) [5]:

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|} = \frac{2TP + \epsilon}{2TP + FP + FN + \epsilon} \quad (3.5)$$

Additionally, the Precision and Recall are reported to differentiate between the models' predictive accuracy and its ability to identify all relevant target pixels [5]:

- **Precision:** The ratio of correctly predicted positive pixels to the total predicted positive pixels

$$Precision = \frac{TP + \epsilon}{TP + FP + \epsilon} \quad (3.6)$$

- **Recall:** The ratio of correctly predicted positive pixels to all pixels in the ground truth.

$$Recall = \frac{TP + \epsilon}{TP + FN + \epsilon} \quad (3.7)$$

Where TP (True Positives), FP (False Positives), and FN (False Negatives) are calculated by comparing the binarized model output (at a threshold of 0.5) against the ground truth mask.

To provide a more tangible measurement of spatial error, the 95th Percentile Hausdorff Distance (HD95) is used. HD95 measures the maximum of the minimum distances between the boundaries of the predicted and ground-truth segmentations, specifically at the 95th percentile to mitigate the influence of small outlier errors.

Finally, the Convergence Rate is recorded as the number of epochs required for the validation loss to reach stability. This metric is used to evaluate the computational efficiency and training dynamics of various optimizer and loss function configurations.

3.5 Chapter Summary

The individual components of the Methodology are summarized in Figure 3.1 to visualize the end-to-end flow of training and evaluating each model. 5-Fold training for each 'configuration' results in 40 training runs for the 8 classification experiments in Table 3.1 per modality and 135 for the 27 segmentation experiments in Table A.1.

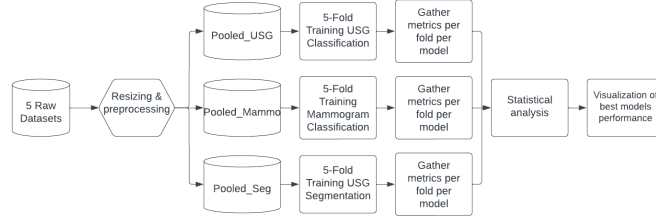


Figure 3.1: Conceptual Training Pipeline

This chapter has detailed the systematic framework designed to evaluate deep learning performance across multiple breast imaging modalities and tasks. By establishing a robust preprocessing pipeline, including inpainting, ROI cropping, and deduplication, the data was conditioned to ensure that model learning is driven by pathological features rather than clinical artifacts or noise. The adoption of a stratified 5-fold cross-validation strategy, paired with patient-level splitting, provides a rigorous defense against data leakage and ensures that the reported metrics ($F1$, AUC , DSC , and $HD95$) are statistically significant. Furthermore, the design of the three ablation studies provides a granular mechanism to isolate the impact of architectural inductive biases (CNNs vs. Transformers) and objective functions (Focal vs. Dice Loss) on diagnostic accuracy. With the research design and experimental parameters firmly established, the following Chapter, 4, will detail the software architecture developed to realize this pipeline, the specific code structure utilized for the ablation studies, and the technical hurdles overcome during the training and deployment phases.

4 Implementation

This chapter describes the implementation of the separate Classification and Segmentation workstreams, highlighting their similarities and key differences, the development process and its iterative improvements, along with justifying design choices. Finally the section concludes with a detailed description of the model orchestration which enables the scalability of this solution.

The implementation phase was conceptualized as a modular and performant Python-based ecosystem. Unlike monolithic "black-box" scripts, this architecture was designed to decouple data engineering from model orchestration. This ensures that the same pre-processing logic is applied consistently across both classification and segmentation workstreams, mitigating "silent failures" where data is handled inconsistently between disparate experiments.

4.1 Data Pipeline Implementation

The data pipeline was implemented to bridge the gap between raw medical data and "model-ready" tensors. This involved two stages: in the first stage raw data was indexed and metadata such as the presence of a valid lesion mask, label, data source, and calculated DHash was gathered. This metadata was used to pool the raw data into the pooled datasets presented in Section 3.2. The second stage is performed in-flight during training and involves the application of specific 'transforms' from the *Albumentations* library in a composition to simulate real-world distractions for the model and increase generalization. Finally the images are transformed to PyTorch $channels \times height \times width$ tensors, the data type expected by the pre-trained models described below. This approach provides a distinct advantage for the training process: by separating data ingestion from final augmentations various ablations can be performed on the same pooled data with different levels of processing. Additionally, the pooled datasets can be version-controlled to ensure distributed learning is using comparable data.

A number of important challenges that should be addressed in the pipeline are outlined below.

Preprocessing and Artifact Removal. Medical imaging datasets often contain "burned-in" annotations or technician artifacts that can lead to shortcut learning, where the model learns to identify the annotation rather than the pathology. An Automated Inpainting Module was implemented using the Telea algorithm via OpenCV. This module masks high-intensity pixels (annotations) and reconstructs the underlying tissue structure based on neighborhood pixel gradients.

Concurrency and Performance. Given the volume of high-resolution images, sequential processing was identified as a bottleneck. The first stage of the data processing pipeline was thoroughly parallelized using the Python *ThreadPoolExecutor*. Decoupling I/O operations from CPU-intensive CLAHE transformations reduced the processing time for the 3,000+ image dataset by approximately 98% (from over 3 hours to under 2 minutes).

Metadata and Stratification. During the training phase, *scikit-learn's StratifiedKFold* was used to read the appropriate metadata and ensure that the 5-fold splits were balanced by class, or lesions size in the case of segmentation, and strictly partitioned by patient across all ablation runs thanks to the thorough tracking for every image.

4.1.1 Dataset Classes

Two classes extended the PyTorch Dataset class, for Classification and Segmentation respectively. They were responsible for the second stage of the pipeline: loading pre-processed images, applying final visual transformations, and directly used by the DataLoaders of the training scripts during training.

Classification Dataset. The class supported customizable configurations of Region of Interest (ROI) cropping, resizing, and 2 levels of augmentation of training data. For baseline training no new augmentation was applied and the saved images were returned as normalized tensors. If augmentation was applied it consisted of an Albumentations Compose containing: a randomly selected one of sharpening or Contrast Limited Adaptive Histogram Equalization (CLAHE), a technique used to increase contrast, random resized cropping, horizontal and vertical flips, random brightness adjustments, random rotations up to 20 degrees, another one of composition of different types of dropouts (where small squares of the image are blacked out), and a final one of composition of blurring or image compression.

Alternative augmentations such as Color Jittering or Random Erasing were avoided due to being non-representative of monochromatic medical image data or would potentially remove diagnostic data required for an accurate prediction. This exhaustive list of techniques was used to stretch the dataset to its absolute maximum in terms of generalization to address the relatively low number of images.

Segmentation Dataset. In addition to its own set of Albumentations transforms, the class was also responsible for ensuring masks matched up to their respective images to ensure a valid training context. The segmentation ablations required 3 separate levels of augmentation, starting from none for a baseline study, increasing up to a very aggressive strategy which involved multiple translations and image manipulation techniques, available in Appendix A.2.

4.2 Classification Framework

Python Image Models (timm). The backbone of the classification framework was the *timm* library [38]. It is a collection of SOTA computer vision models trained on ImageNet-1k [39], a database of over 14 million images. *timm* provides an easy to use robust API which returns ready-to-fine-tune models and is highly configurable.

Warm-Up Training Logic. Since the pretrained models were already cap-

4 Implementation

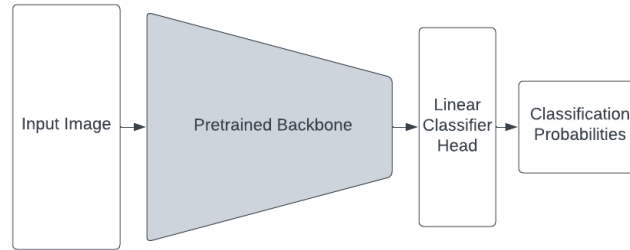


Figure 4.1: Sample Classification Architecture Diagram. Frozen Backbone indicated in gray preserves ImageNet weights while classification head is trained.

able of achieving SOTA performance on general computer vision tasks, a two-stage training process with an initially frozen backbone was implemented to preserve the backbone weights, as seen in Figure 4.1. By setting the backbone parameters *requires_grad = False* only the custom classification head trained for 5 epochs to align the random weights of the head with the pre-trained features. This allows the model to preserve its ImageNet training while learning to recognize modality-specific features in the training data.

4.3 Segmentation Framework

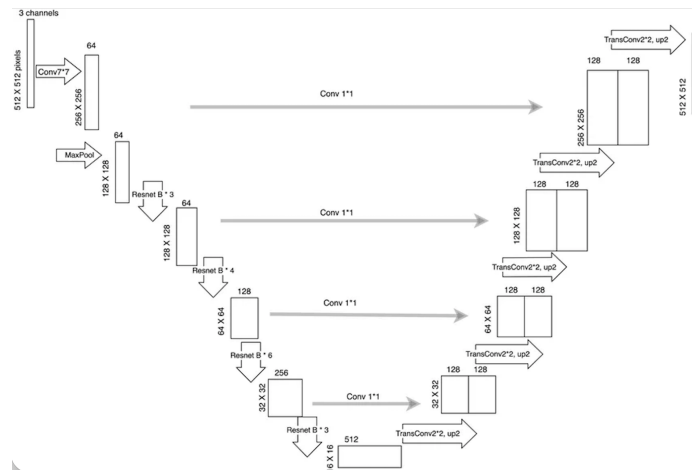


Figure 4.2: Sample ResNet34-UNet Architecture Diagram[40]. ResNet (the encoder) downsamples the image while UNet upsamples it.

Segmentation-Models-Pytorch (SMP). The backbone of the segmentation framework was the SMP library which provides pre-trained segmentation models, using ImageNet [39] weights and was highly customizable for various encoder-decoder architectures. Figure 4.2 visualizes a sample model architecture used in this study. The model takes in a fixed-size image

and returns a binarized mask of the identified lesion. Due to the larger size of the segmentation models, a smaller image size of 512×512 was used due to hardware constraints.

4.4 Framework Orchestration

The guiding thought behind the design of each component in the pipeline was the ability to scale the training process to multiple computational devices seamlessly, therefore a Functional Modular pattern was adopted for the orchestration logic. All ablations were translated to YAML configuration files which were used by a central delegate to set up correct models, optimizers, loss functions, and training modes (SWA or early stopping). In addition, the pipeline adapted to the hardware environment and could run on both Apple's MPS and NVIDIA's CUDA thanks to a hardware abstraction layer implemented using PyTorch's backend detection.

Each framework returned a set of files per ablation: a csv file containing all metrics gathered throughout the training run and a log file containing metadata information about the training process such as the convergence rate and training times.

This allowed for seamless transitions between both environments to ensure rapid prototyping and debugging, especially when running the 200+ ablations required, and an easy experience combining data from multiple devices.

4.5 Technical Challenges and Solutions

Developing a SOTA pipeline on consumer hardware presented several interesting technical challenges.

Memory Management: The 8GB of VRAM in the RTX 3070 Ti Laptop GPU was insufficient for training high-resolution Transformers at the required batch size of 16. Implementing Gradient Accumulation mitigated this by computing gradients over smaller 'mini-batches' of 8 and updating the weights over 2 steps, effectively simulating a larger batch size. Even with this advanced solution loading 1024×1024 segmentation images and masks onto desktop GPUs with 24GB proved too challenging resulting in a necessary modification to the scope of the study.

Mixed Precision Training (AMP): To further optimize memory management and speed Automatic Mixed Precision was utilized via `torch.amp.cuda`. By performing forward passes using half precision floats (F16) while maintaining a copy of F32 weights, the memory footprint was halved, and training speed increased noticeably from 7 minutes per epoch to 5.

MPS vs CUDA Parity: A significant hurdle was that certain operations (like specific types of 2D Interpolation or complex loss functions) behaved differently or were unsupported on Apple's relatively-newer MPS. Therefore, custom wrappers were implemented to use device-compatible implementations, falling back on the CPU for unsupported operations to ensure identical results cross-platform.

Checkpointing: On the M4 Pro chip the unified memory is used by both the MPS and CPU which meant that keeping large objects in memory over the multiple-hour training sessions eroded performance over time. To overcome this limitation, model weights were saved to files after each fold and loaded back when required for analytics to preserve memory.

4.6 Chapter Summary

This chapter outlines the overall implementation strategy of the project, beginning with data preparation and augmentation to maximize learnable features from a small dataset to the implementation of Classification and Segmentation Frameworks for reproducible model training across hundreds of repetitions and accurate metrics gathering. The orchestration of all modular components is discussed and rationalized, including the selection of hyperparameters and how ablations are evaluated. The resulting software suite provides a robust, end-to-end framework for breast cancer imaging research.

5 Results and Evaluation

5.1 Classification Results and Evaluation

This section analyzes four architectures across Mammography and USG modalities using 5-fold cross-validation.

5.1.1 Architecture Benchmarking

Initial baselines (Table 5.1) confirm a consistent "**Modality Gap**," where USG performance significantly exceeds Mammography. ConvNeXt Small achieved an F1 of 0.7909 on USG versus 0.7149 on Mammography. This gap is attributed to:

- **Acquisition Physics:** Mammography's 2D projection causes "tissue masking," whereas USG's cross-sectional nature provides a higher signal-to-clutter ratio.

- **Feature Scale:** Mammographic microcalcifications are high-frequency signals easily lost during downsampling; USG features (e.g., irregular margins) are more robust to spatial compression.
- **Contrast Gradients:** USG exhibits sharp lesion-to-stroma contrast, while Mammography often presents "isodense" lesions that share grayscale intensities with healthy tissue.

The Transformer Anomaly: Swin-Transformer peaked at 0.7947 (USG) but showed high variance and lower mammographic stability (0.7070), suggesting sensitivity to high-noise distributions.

Notably, a potential outlier of 0.9853 AUC (ConvNeXt-Mammo) was identified as a non-reproducible artifact of overfitting. Following a data audit, a conservative baseline of 0.7088 was adopted to ensure clinical reliability.

Table 5.1: Architecture Classification Benchmark

Architecture	F1-Score ($\pm\sigma$)	AUC-ROC	Modality
No Augmentation ConvNeXt	0.6995 $\pm$ 0.020	0.8605	Mammogram
No Augmentation ConvNeXt	0.7748 $\pm$ 0.019	0.8978	USG
ConvNeXt Small	0.7149 $\pm$ 0.005	0.8799	Mammogram
ConvNeXt Small	0.7909 $\pm$ 0.012	0.9109	USG
EfficientNet Small	0.6735 $\pm$ 0.028	0.8399	Mammogram
EfficientNet Small	0.7571 $\pm$ 0.014	0.8937	USG
Hybrid Max ViT	0.5819 $\pm$ 0.093	0.7905	Mammogram
Hybrid Max ViT	0.7887 $\pm$ 0.006	0.9160	USG
ViT Swin Tiny	0.7070 $\pm$ 0.018	0.8746	Mammogram
ViT Swin Tiny	0.7947 $\pm$ 0.015	0.9163	USG

5.1.2 Ablation Analysis: ROI, Focal Loss, and SWA

Ablation results (Table 5.2) demonstrate the impact of clinical preprocessing and optimization:

- **ROI Cropping:** Provided a universal "lift" (USG: 0.8035, $\sigma = 0.002$), stabilizing model focus by removing non-diagnostic background.
- **Focal Loss:** Caused a marginal F1 drop (0.7254 $\rightarrow$ 0.7148 in Mammo) but increased sensitivity to "hard" malignant samples, a critical trade-off for clinical screening.
- **SWA Paradox:** SWA improved CNN generalization by finding flatter minima. However, Swin-Transformer performance collapsed (0.7921 $\rightarrow$ 0.7019). This suggests weight averaging "blurs" the sharp,

high-frequency attention maps Transformers require for high-noise USG features.

Table 5.2: Classification Results

Experiment	F1-Score ($\pm\sigma$)	AUC-ROC	Modality
ConvNeXt Small + ROI	0.7254 ± 0.024	0.8848	Mammogram
ConvNeXt Small + ROI	0.8035 ± 0.002	0.9142	USG
+ Focal Loss	0.7148 ± 0.011	0.8776	Mammogram
+ Focal Loss	0.7905 ± 0.017	0.9102	USG
+ SWA	0.7088 ± 0.018	0.8764	Mammogram
+ SWA	0.7977 ± 0.011	0.9201	USG
ViT Swin Tiny + ROI + Focal Loss	0.7123 ± 0.024	0.8743	Mammogram
ViT Swin Tiny + ROI + Focal Loss	0.7921 ± 0.008	0.9197	USG
+ SWA	0.6987 ± 0.030	0.8657	Mammogram
+ SWA	0.7019 ± 0.015	0.8874	USG
EfficientNet Small + ROI + Focal Loss	0.6702 ± 0.009	0.8482	Mammogram
EfficientNet Small + ROI + Focal Loss	0.7606 ± 0.011	0.9005	USG
Hybrid Max ViT + ROI + Focal Loss	0.6702 ± 0.012	0.8439	Mammogram
Hybrid Max ViT + ROI + Focal Loss	0.7936 ± 0.015	0.9155	USG

5.1.3 Statistical Significance

Hedge’s g effect sizes confirm these findings. In USG (Appendix C.1), ROI Cropping showed a robust $g = 1.15$, while the SWA-Transformer conflict yielded a catastrophic $g = -4.88$. In Mammography (Appendix C.2), the Final Model (ROI+Focal+SWA) achieved a massive $g = 14.94$ over baseline, proving that integrated preprocessing and smoothed optimization are essential for navigating the low-contrast mammographic environment.

While individual techniques like ROI cropping provided moderate improvements on their own ($g = 0.49$ over baseline), their integrated application yielded a model that is statistically superior to any single-modality baseline. Collectively, these g -values heatmaps validate the research methodology.

5.1.4 Interpretability and Training Dynamics

Grad-CAM analysis (Figures 5.1, 5.2) illustrates that **ConvNeXt** acts as a "High-Frequency" detector (localized, sharp activations), while **Swin-Transformer** acts as a "Structural" detector (diffuse, context-aware activations). This provides a qualitative "sanity check" to ensure that the models are focusing on pathological features rather than background noise. SWA further refined ConvNeXt focus while visibly degrading Swin’s attention maps.

In Figure 5.1a, the ConvNeXt + ROI Cropping model successfully localizes a suspicious mass in a mammogram. The heatmap exhibits a concentrated,

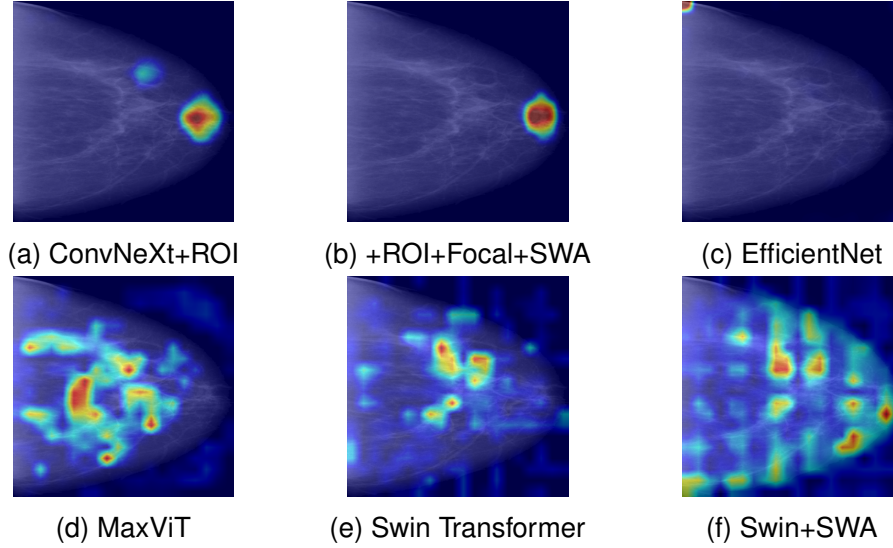


Figure 5.1: Comparison of Mammography Models Across Same Image. ConvNeXt demonstrates highly localized attention points.

"peak-like" activation directly over the lesion. This demonstrates that the local inductive bias of the CNN, when coupled with the automated ROI cropping implemented in this study, effectively filters out glandular density noise. Clinically, this suggests that the model is making decisions based on relevant pathological textures, specifically the architectural distortion of the tumor, rather than extraneous anatomical markers or technician artifacts.

Training trajectories confirmed stability: models achieved 80% of peak performance within 10 epochs due to the 5-epoch warm-up and transfer learning (Appendices C.3 and C.4). A 12-epoch patience was justified as F1-scores trended upward until Epoch 40. The stochastic fluctuations post-Epoch 20 (Figure C.3a) further justify the use of SWA to reach a non-oscillating, generalized solution.

5.2 Segmentation Results and Evaluation

5.2.1 Architectural and Optimizer Benchmarking

Initial benchmarking (Table 5.3) demonstrates that while ResNet34 + UNet++ provides balanced precision, the MiT-B3 encoder achieves superior recall (0.8709). However, MiT-B3 exhibited a higher Hausdorff Distance (HD95), indicating less boundary precision in baseline configurations.

A pivotal finding was the impact of optimizer "Simplicity" (Table 5.4). While uniform learning rates ("Simple") generally degraded ResNet performance, they consistently improved MiT-B3 across all loss functions, significantly reducing variance. The MiT-B3 Adam Simple Focal model emerged as

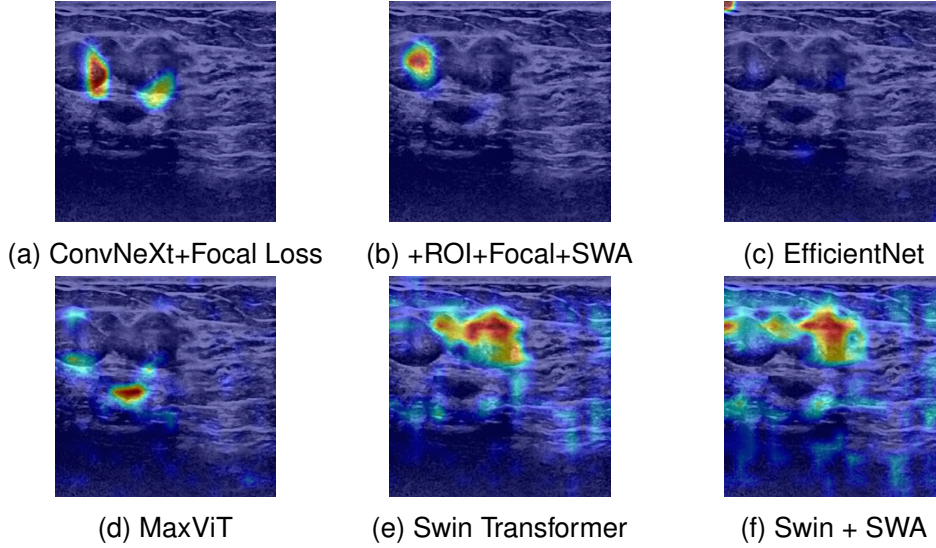


Figure 5.2: Comparison of Ultrasound Models Across Same Image. Impact of SWA is visible in model activations.

Table 5.3: Architecture Comparison: Tversky Loss Function, Optimizer AdamW with Differential LRs, image size 512×512

Encoder + Decoder	Dice Score $\pm\sigma$	Precision $\pm\sigma$	Recall $\pm\sigma$	HD95 $\pm\sigma$ (mm)
ResNet34 + UNet++	0.7091 $\pm$ 0.08	0.7405 $\pm$ 0.1072	0.8235 $\pm$ 0.0218	39.08 $\pm$ 4.72
ResNet34 + UNet	0.6789 $\pm$ 0.06	0.6674 $\pm$ 0.1121	0.8566 $\pm$ 0.441	41.49 $\pm$ 12.03
MiT-B3 + UNet	0.6866 $\pm$ 0.08	0.6700 $\pm$ 0.1155	0.8709 $\pm$ 0.04	46.94 $\pm$ 21.03

the performance frontier, achieving the highest Dice score (0.7733) and demonstrating that Transformer-based encoders benefit from more stable, less aggressive learning schedules.

5.2.2 Impact of Image Size and Preprocessing

Ablations on input resolution (Table 5.5) reveal a "localization collapse" at low resolutions. While 224×224 inputs yielded competitive Dice scores, HD95 values exploded (> 130 mm), indicating a loss of boundary fidelity. Increasing resolution to 1024 improved spatial localization but offered diminishing returns for region-based metrics (Dice).

Preprocessing experiments (Table 5.5) confirmed that while basic cleaning is essential (preventing the 0.58 Dice baseline seen in "None"), Aggressive preprocessing caused a uniform performance drop. This suggests that over-cleaning removes vital contextual cues required for accurate lesion delineation. A moderate preprocessing strategy is therefore required to balance noise removal with feature preservation.

Table 5.4: Criterion and Optimizer Comparison: ResNet Decoder UNet++, MiT-B3 UNet, image size 512×512

Encoder, Optimizer, Criterion	Dice Score $\pm\sigma$	Precision $\pm\sigma$	Recall $\pm\sigma$	HD95 $\pm\sigma$ (mm)
ResNet Adam Tversky	0.7091 $\pm$ 0.0763	0.7405 $\pm$ 0.1072	0.8235 $\pm$ 0.0218	39.08 $\pm$ 4.72
ResNet Adam Simple Tversky	0.6998 $\pm$ 0.0201	0.6975 $\pm$ 0.0291	0.8336 $\pm$ 0.0210	40.25 $\pm$ 12.71
ResNet Adam BCE	0.7354 $\pm$ 0.0507	0.7986 $\pm$ 0.0538	0.7854 $\pm$ 0.0274	28.63 $\pm$ 10.42
ResNet Adam Simple BCE	0.7129 $\pm$ 0.0284	0.7825 $\pm$ 0.0276	0.7725 $\pm$ 0.0289	35.71 $\pm$ 9.21
ResNet Adam 0.2 BCE	0.7497 $\pm$ 0.0320	0.7675 $\pm$ 0.0418	0.8375 $\pm$ 0.0214	28.94 $\pm$ 8.68
ResNet Adam Simple 0.2 BCE	0.6736 $\pm$ 0.0344	0.6426 $\pm$ 0.0496	0.8668 $\pm$ 0.0223	38.33 $\pm$ 6.62
ResNet Adam 0.3 BCE	0.6931 $\pm$ 0.0600	0.6653 $\pm$ 0.0829	0.8638 $\pm$ 0.0392	36.92 $\pm$ 10.95
ResNet Adam Simple 0.3 BCE	0.6649 $\pm$ 0.0474	0.6307 $\pm$ 0.0627	0.8558 $\pm$ 0.0240	42.46 $\pm$ 13.98
ResNet Adam Focal	0.7376 $\pm$ 0.0125	0.8273 $\pm$ 0.0313	0.7778 $\pm$ 0.0177	32.12 $\pm$ 9.71
ResNet Adam Simple Focal	0.7111 $\pm$ 0.0365	0.7656 $\pm$ 0.0769	0.7886 $\pm$ 0.0458	36.77 $\pm$ 11.52
MiT-B3 Adam Focal	0.7541 $\pm$ 0.0357	0.7712 $\pm$ 0.0405	0.8462 $\pm$ 0.0159	35.95 $\pm$ 19.69
MiT-B3 Adam Simple Focal	0.7733 $\pm$ 0.0254	0.8053 $\pm$ 0.0300	0.8308 $\pm$ 0.0248	36.88 $\pm$ 18.49
MiT-B3 Adam BCE	0.6966 $\pm$ 0.0477	0.7088 $\pm$ 0.0618	0.8308 $\pm$ 0.0291	41.25 $\pm$ 19.58
MiT-B3 Adam Simple BCE	0.7637 $\pm$ 0.0203	0.7994 $\pm$ 0.0156	0.8263 $\pm$ 0.0332	42.40 $\pm$ 21.45
MiT-B3 Adam 0.3 BCE	0.7411 $\pm$ 0.0367	0.7331 $\pm$ 0.0398	0.8615 $\pm$ 0.0241	32.49 $\pm$ 14.12
MiT-B3 Adam Simple 0.3 BCE	0.7406 $\pm$ 0.0233	0.7109 $\pm$ 0.0288	0.8848 $\pm$ 0.0199	43.78 $\pm$ 15.87
MiT-B3 Adam 0.2 BCE	0.7465 $\pm$ 0.0543	0.7405 $\pm$ 0.0824	0.8589 $\pm$ 0.0285	37.93 $\pm$ 15.18
MiT-B3 Adam Simple 0.2 BCE	0.7500 $\pm$ 0.0274	0.7364 $\pm$ 0.0404	0.8708 $\pm$ 0.0203	35.61 $\pm$ 15.43
MiT-B3 Adam Tversky	0.6866 $\pm$ 0.0758	0.6700 $\pm$ 0.1155	0.8709 $\pm$ 0.0435	46.94 $\pm$ 21.03
MiT-B3 Adam Simple Tversky	0.7595 $\pm$ 0.0237	0.7427 $\pm$ 0.0325	0.8711 $\pm$ 0.0206	37.96 $\pm$ 17.45

Table 5.5: Resolution and Preprocessing Impact

Variable	Config	Dice Score	Recall	HD95 (mm)
Resolution	MiT-B3 (224px)	0.7307	0.8592	139.93 $\pm$ 19.1
Resolution	ResNet (1024px)	0.6974	0.8077	93.73 $\pm$ 20.2
Preprocessing	None (ResNet)	0.5808	0.6984	34.77 $\pm$ 21.9
Preprocessing	Aggressive (MiT)	0.6736	0.8793	51.37 $\pm$ 13.4

5.2.3 Statistical Significance and Visualization

Hedge’s g effect sizes (Appendix D.1) validate the superiority of the Transformer-based approach. The MiT-B3 Simple Focal model demonstrated a large effect size over its ResNet counterpart ($g = 1.60$) and a massive gain over standard 224px ResNet baselines ($g = 2.15$). In the MiT-B3 framework, the transition from BCE to Focal loss yielded a substantial positive effect ($g = 1.08$), confirming that Focal loss effectively manages the foreground-background class imbalance inherent in USG segmentation.

The benchmarking results collectively demonstrate a clear paradigm shift from traditional CNN-based architectures to transformer-based encoders in this segmentation task. While ResNet-34 remains a robust baseline for precision-oriented tasks, the MiT-B3 encoder coupled with a Focal loss function and a simplified optimization strategy represents the current performance frontier. The large effect sizes ($g > 1.0$) observed when transitioning to "Simple" MiT-B3 variants suggest that transformer-based models

5 Results and Evaluation

may benefit from more stable, less aggressive learning rate schedules than their convolutional counterparts. Furthermore, the decoupling of the Dice score and HD95 in low-resolution experiments highlights that while global segmentation remains feasible at 224px, high-resolution inputs are non-negotiable for applications requiring boundary fidelity.

Qualitative analysis (Figure 5.3) reinforces these metrics. High-performing models successfully localized both the smallest and largest lesions, despite the "pixel-imprecision" of the technician-drawn Ground Truth. The ResNet Aggressive model's failure to identify Sample 0 visually confirms the quantitative "information loss" caused by over-augmentation. Collectively, these results mark a paradigm shift: while ResNet remains robust for precision-heavy tasks, the MiT-B3 encoder, paired with simplified optimization and Focal loss, represents the new SOTA for USG lesion delineation. A full comparison of all models is available in Figure D.4.



Figure 5.3: Sample model performance across differently sized lesions. GT is the technician-drawn mask for the image.

5.3 Evaluation Against Literature

USG Segmentation and Classification. The best performing segmentation model in this study, the MiT-B3 encoder utilizing a uniform learning rate AdamW optimizer and Focal loss, was evaluated on the BUSI dataset [13]. The model achieved a reproducible average Dice Score of 0.90 and a Precision of 0.92. This performance aligns closely with current State-of-the-Art (SOTA) in modified Transformer-based architectures, which typically range between 0.90 and 0.92 (e.g., Attention-Driven UNet [41] and FuseMix-BEAM [42]). This suggests that the ablation methodology employed here effectively reached SOTA-level parity in an explainable manner.

Similarly, the ConvNeXt USG classification model (utilizing BCE loss and ROI cropping) demonstrated excellent diagnostic capability with an F1 Score of 0.96 and an AUC of 0.9890 on BUSI. As evidenced by the confusion matrix in Appendix C.5, the model maintained high sensitivity across both benign and malignant classes, with minimal cross-class leakage. While

this performance is slightly below architectural benchmarks in other high-complexity medical tasks, such as Alzheimer’s detection (F1 0.9837, AUC 0.992 [43]), it confirms the model’s high sensitivity to the morphological features present in ultrasound imaging.

Mammography Classification In contrast, the Mammography classification model (ConvNeXt with Focal loss and SWA) achieved an AUC of 0.68 on the INBreast dataset [31], falling short of the 0.87 AUC reported in recent literature [44]. As illustrated in the confusion matrix (Appendix C.6), the model struggled significantly with class-to-class discrimination outside of the majority class.

Specifically, the model exhibited a high degree of confusion between ‘Benign’ and ‘Normal’ categories, with 25.4% (73/287) of benign cases misclassified as normal. This suggests that the global feature extraction of the ConvNeXt backbone was unable to sufficiently distinguish between benign architectural changes and normal glandular density.

Furthermore, the model yielded a 36% False Negative rate for malignant cases (36/100). This performance gap can be attributed to several factors:

- **Feature Resolution and Global vs. Local Context:** Unlike USG images, which focus on localized regions of interest, mammograms are high-resolution, large-format images where malignant indicators, such as microcalcifications or subtle architectural distortions, occupy a fraction of the total pixel area. The ConvNeXt architecture, while robust, may require more specialized "Global-Local" attention mechanisms to capture these nuances compared to the more distinct mass boundaries found in USG.
- **Data Heterogeneity and Pooling:** The private datasets integrated into this study likely introduced higher variance in terms of acquisition hardware and compression artifacts compared to the curated INBreast benchmark.

The comparatively higher results across both USG tasks suggest that the data pooling methodology successfully created a "stress-test" environment. The drop in mammography performance indicates that the model was forced to contend with a more authentic, noisy, and challenging clinical distribution than that found in isolated public benchmarks.

5.4 Chapter Summary

This chapter has quantified the performance frontiers of deep learning in breast imaging, identifying ultrasound as the more robust modality for automated feature extraction. The findings established that while CNNs excel at

capturing local pathological textures, Transformers provide superior global structural context, though they remain sensitive to specific optimization techniques like SWA. The systematic benchmarking of segmentation work streams confirmed the superiority of the MiT-B3 encoder under uniform optimization schedules and highlighted high-resolution input as a prerequisite for spatial boundary precision. Collectively, these results provide a statistically validated framework for selecting optimal architectures and objective functions for clinically trustworthy computer-aided diagnosis.

6 Conclusion

This research has provided a broad-reaching and multi-faceted evaluation of deep learning architectures for breast cancer diagnosis, successfully bridging the research gap between traditional convolutional approaches and emerging transformer-based paradigms across multi-modal imaging. By implementing a unified experimental framework, this study has not only identified the performance frontiers of modern CAD systems but has also established the technical and clinical prerequisites for their deployment.

6.1 Synthesis of Outcomes and Success Criteria

The project successfully met its self-defined success criteria, utilizing 5-fold stratified cross-validation to ensure that all reported metrics, F1-score, AUC-ROC, and Dice Similarity Coefficient (DSC), possess statistical significance and are not artifacts of data splits.

- **Classification Outcomes:** The investigation identified a persistent "Modality Gap," where ultrasound (USG) consistently yielded higher diagnostic accuracy (F1: 0.7909) than mammography (F1: 0.7149). This outcome proves that while mammography remains the "gold standard," ultrasound provides more distinct morphological features for automated filters, fulfilling the project's aim to address the operator-dependency of USG through robust automation.
- **Segmentation Outcomes:** The MiT-B3 encoder coupled with Focal Loss emerged as the performance frontier, achieving a peak Dice score of 0.7733. This result, validated by Hedge's g effect sizes ($g > 1.0$), confirms that Transformer-based encoders benefit from more stable, uniform learning rate schedules than their convolutional counterparts, representing a significant outcome for future architectural design.

6.2 Evaluation against Original Aims and Motivation

This study answered all 4 Research Questions posed in Chapter 2:

- **Regarding the Modality Gap (RQ1):** The project identified a consistent "Modality Gap", where USG images yielded higher F-1 scores (0.7909) compared to mammography (0.7149). This suggests that USG provides more distinct morphological features for automated filters, despite mammography being the clinical "gold standard".
- **Regarding Preprocessing and Shortcut Learning (RQ2):** ROI Cropping provided a universal "lift" and was essential for clinical integrity. By implementing the Telea algorithm for inpainting, the model was not able to "cheat" by identifying burned-in annotations rather than underlying pathology.
- **Regarding Architectural Inductive Biases (RQ3):** The research provides a complex answer to the CNN vs Transformer debate. CNNs function as "High-Frequency" detectors that benefit from the smoothing effect of SWA. In contrast, Transformers contextualize the lesion within surrounding tissue, thereby acting as "Structural" detectors, but are sensitive to SWA which "blurs" their highly specific attention maps.
- **Regarding the Segmentation Frontier (RQ4):** The MiT-B3 Transformer Encoder coupled with Focal Loss and a AdamW uniform learning rate optimizer represents the current performance frontier. The very large effect sizes ($g > 1.0$) observed when using uniform optimizers suggest that Transformer-based segmentors benefit from more stable training schedules than traditional convolutional models.

While this project has successfully answered RQ1-RQ4, the identified limitations regarding the Transformer-SWA conflict and boundary fidelity (HD95) have established a clear roadmap for future hybrid architectural research and multi-modal fusion systems.

6.3 Substantial Implications and Clinical Trustworthiness

The implications of this research extend beyond numerical accuracy to the qualitative interpretability of CAD systems.

- **Architectural Inductive Biases:** The discovery that CNNs act as "High-Frequency" detectors (sharp edges) while Transformers act as

"Structural" detectors (global context) has profound implications for clinician trust. The Grad-CAM analysis argued that models with concentrated, localized heatmaps are more "trustworthy" for a radiologist than those distracted by high-contrast edges.

- **Technical Viability:** By overcoming hardware constraints through Gradient Accumulation and Automatic Mixed Precision (AMP), this study demonstrates that SOTA diagnostic performance is achievable on consumer-grade hardware, significantly lowering the barrier for real-world clinical deployment.

6.4 Future Work

Classification. To resolve the identified "Modality Gap" (where USG consistently outperformed mammography by an F1 margin of 0.076), Dual-Branch Cross-Attention architecture is proposed. This design directly addresses the discovered architectural dichotomy: ConvNeXt as a "High-Frequency" detector for subtle calcifications and Swin-Transformers as "Structural" detectors for morphological context.

Technical Design:

- **CNN-Mammography Branch:** Utilizes a ConvNeXt-Small encoder to extract high-resolution, local features. As established in this study, CNNs are superior at capturing the sharp edges of microcalcifications that mammography is clinically prioritized for.
- **ViT-Ultrasound Branch:** Utilizes a Swin-Transformer encoder to capture global spatial dependencies. This leverages the Transformer's ability to contextualise lesions within surrounding healthy tissue, mitigating the "operator-dependency" artifacts common in USG.
- **Cross-Attention Fusion:** Instead of simple concatenation, a Cross-Attention module should be implemented between the bottleneck layers. By using the USG feature maps as the Query (Q) and the Mammography maps as Key (K) and Value (V), the model can "attend" to mammographic textures only when they align with the structural evidence found in ultrasound.

This fusion directly enhances clinical trustworthiness. By integrating both modalities, the system mimics the multi-modal workflow of a radiologist, potentially reducing the legal risks associated with false negatives in dense breast tissue, a known failure point for mammography alone. Such an "autonomous clinical tool" would not replace professional judgment but would provide a robust, statistically-backed second opinion.

Segmentation. Following the benchmarking of the current segmentation frontiers in RQ4, several avenues for enhancement are identified:

- **Hybrid Architectural Evolution (RQ4):** Although the MiT-B3 encoder achieved the highest recall, the high HD95 scores indicate a need for better boundary precision. Future work will investigate hybrid CNN-Transformer architectures (such as TransUNet), which aim to combine the local edge-detection strengths of CNNs with the global context of Transformers to sharpen lesion boundaries.
- **Cross-Domain and Edge Deployment:** To validate the "Simple" AdamW configuration's robustness, future research should evaluate these models on multi-center datasets with varying noise profiles. Furthermore, to make these high-performing models viable for real-time clinical use, research into model quantization and pruning for the MiT-B3 backbone is essential to facilitate deployment on resource-constrained hardware.

6.5 Final Motivation Linkage

Ultimately, this project confirms that while the "Modality Gap" remains a challenge, the integration of hierarchical transformers and clinical preprocessing can provide the high-fidelity spatial localization and categorical accuracy required for effective breast cancer screening. By relating these technical outcomes back to the global challenge of rising cancer incidence, this research provides a robust framework for the next generation of interpretable and accessible Computer-Aided Diagnosis systems.

A Methodology

A.1 Data Availability

Processed data and trained model weights are available at the following repository: Google Drive: Research Data

A.2 Source Code

All code needed to replicate the presented results in this study is available at the following repository: GitHub

A.3 Ablation Design

Table A.1: Systematic Ablation Design for USG Segmentation where Experiment Name indicated Image Size and Level of Preprocessing

Experiment Name	Encoder + Decoder	Loss Function	Optimizer
No Augmentation Baseline	ResNet34 + UNet++	BCE	AdamW Simple
ResNet Adam Simple Baseline	ResNet34 + UNet++	BCE	AdamW Simple
ResNet Adam Simple 0.3 BCE	ResNet34 + UNet++	0.3 BCE + 0.7 Dice	AdamW Simple
ResNet AdamW Simple Tversky	ResNet34 + UNet++	Tversky	AdamW Simple
ResNet Adam Simple 0.2 BCE	ResNet34 + UNet++	0.2 BCE + 0.8 Dice	AdamW Simple
ResNet Adam Simple Focal	ResNet34 + UNet++	0.3 Focal + 0.7 Dice	AdamW Simple
ResNet Adam Baseline	ResNet34 + UNet++	BCE	AdamW
ResNet Adam 0.3 BCE	ResNet34 + UNet++	0.3 BCE + 0.7 Dice	AdamW
ResNet Adam 0.2 BCE	ResNet34 + UNet++	0.2 BCE + 0.8 Dice	AdamW
ResNet Tversky	ResNet34 + UNet++	Tversky	AdamW
ResNet Adam Focal	ResNet34 + UNet++	0.3 Focal + 0.7 Dice	AdamW
ResNet 224	ResNet34 + UNet++	Tversky	AdamW
ResNet 1024	ResNet34 + UNet++	Tversky	AdamW
ResNet Aggressive	ResNet34 + UNet++	Tversky	AdamW
ResNet Unet	ResNet + UNet	Tversky	AdamW
MIT-B3 Adam Simple Baseline	MiT-B3 + UNet	BCE	AdamW Simple
MIT-B3 Adam Simple Tversky	MiT-B3 + UNet	Tversky	AdamW Simple
MIT-B3 Adam Simple 0.2 BCE	MiT-B3 + UNet	0.2 BCE + 0.8 Dice	AdamW Simple
MIT-B3 Adam Simple Focal	MiT-B3 + UNet	0.3 Focal + 0.7 Dice	AdamW Simple
MIT-B3 Adam Simple 0.3 BCE	MiT-B3 + UNet	0.3 BCE + 0.7 Dice	AdamW Simple
MIT-B3 Adam	MiT-B3 + UNet	BCE	AdamW
MIT-B3 Adam Tversky	MiT-B3 + UNet	Tversky	AdamW
MIT-B3 Adam 0.3 BCE	MiT-B3 + UNet	0.3 BCE + 0.7 Dice	AdamW
MIT-B3 Adam Focal	MiT-B3 + UNet	0.3 Focal + 0.7 Dice	AdamW
MIT-B3 Adam 0.2 BCE	MiT-B3 + UNet	0.2 BCE + 0.8 Dice	AdamW
MIT-B3 Aggressive	MiT-B3 + UNet	Tversky	AdamW
MIT-B3 224	MiT-B3 + UNet	Tversky	AdamW

B Implementation

```
def train_one_epoch(model, loader, optimizer, criterion, device, correct, total, scaler=None):
    model.train()
    running_loss = 0.0

    pbar = tqdm(loader, desc="Training")
    for imgs, labels in pbar:
        imgs, labels = imgs.to(device), labels.to(device)
        optimizer.zero_grad()

        if scaler and device.type == 'cuda':
            with torch.autocast('cuda'):
                outputs = model(imgs)
                loss = criterion(outputs, labels)
            scaler.scale(loss).backward()
            scaler.step(optimizer)
            scaler.update()
        else:
            outputs = model(imgs)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()
        running_loss += loss.item() * imgs.size(0)
        correct += (outputs.argmax(1) == labels).sum().item()
        total += labels.size(0)

    return running_loss / total, correct / total
```

Figure B.1: Example of Hardware-Agnostic Implementation supporting both CUDA Acceleration and MPS

C Classification Results

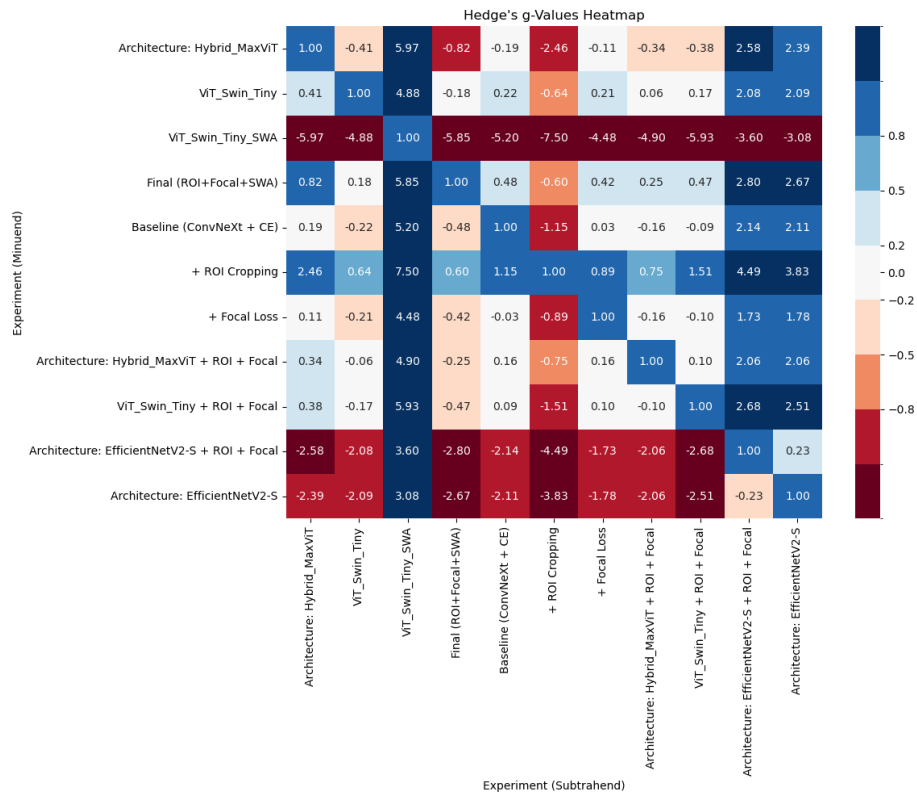


Figure C.1: USG F1 Score Hedge's g-value Heatmap. Results indicate strong significance in the use of ROI Cropping while at the same time highlighting the poor performance of SWA on Transformers

C Classification Results

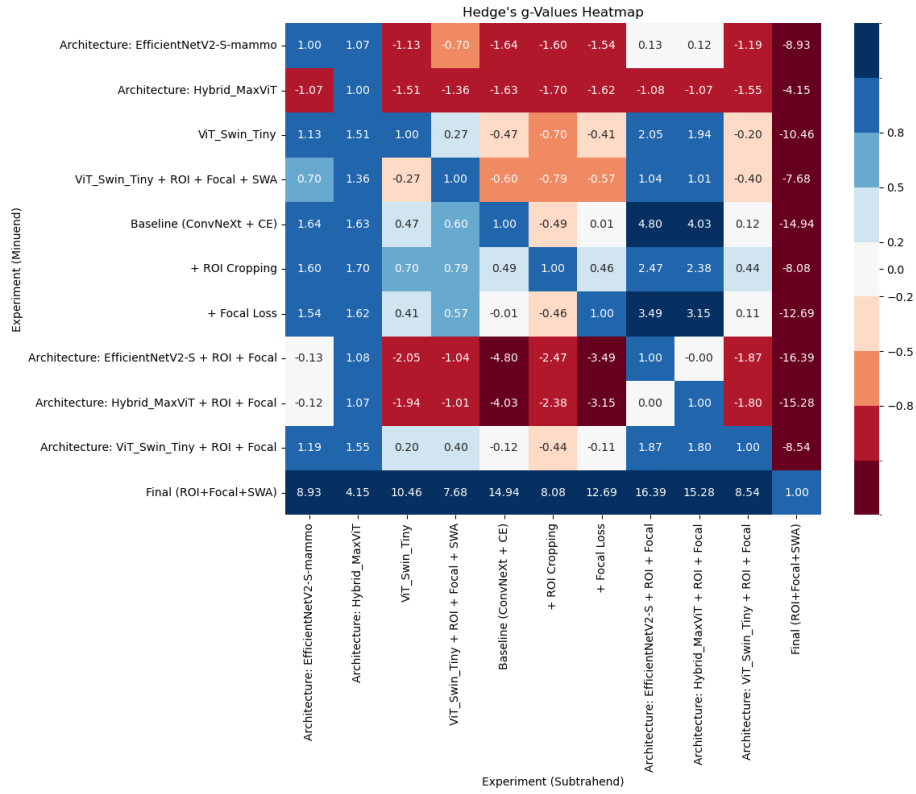


Figure C.2: Mammography F1 Score Hedge's g-value Heatmap. ROI with SWA performs exceptionally well on this modality.



Figure C.3: Training Curves for Models on USG Images

C Classification Results

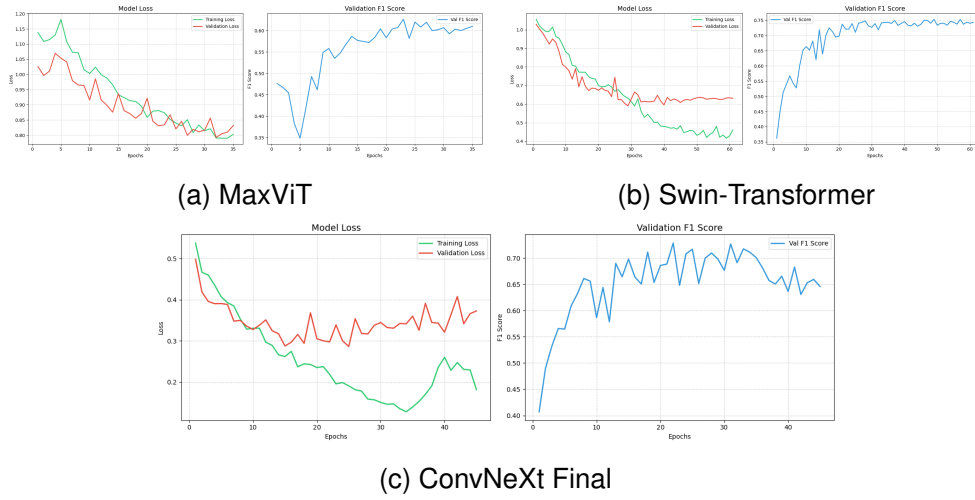


Figure C.4: Training Curves for Models on Mammogram Images

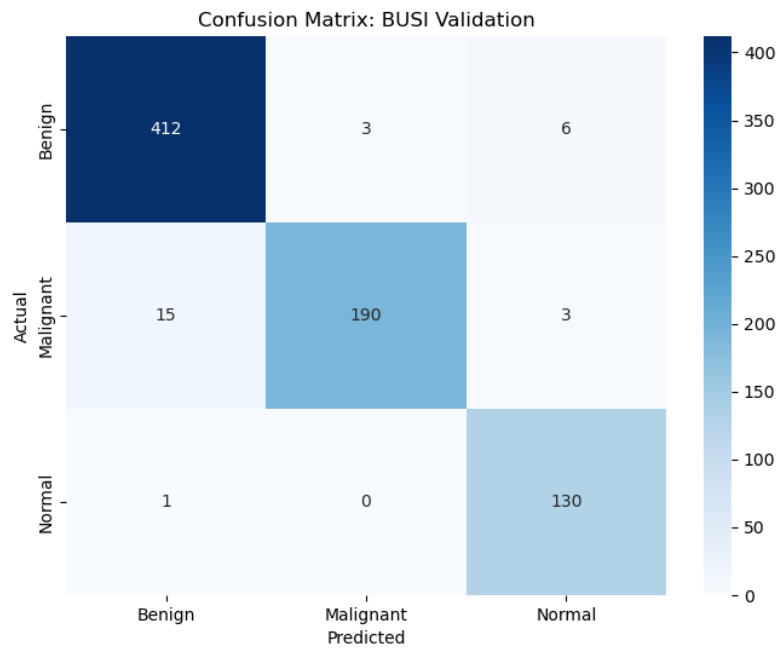


Figure C.5: Confusion Matrix of Best Ultrasound Model on BUSI. 15 Malignant cases identified as Benign provide biggest area of improvement.

C Classification Results

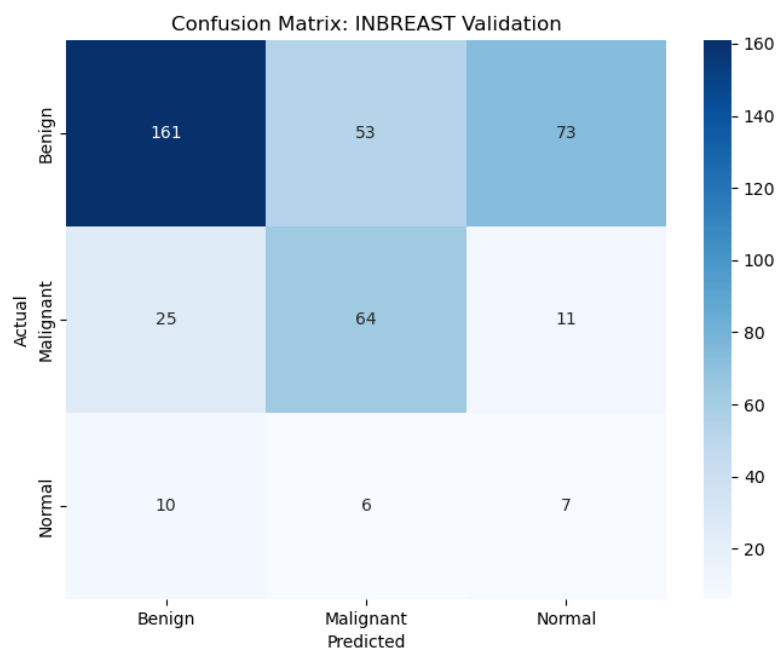


Figure C.6: Confusion Matrix of Best Mammogram Model on INBreast. 36 Malignant cases identified as Benign or Normal provide biggest area of improvement.

D Segmentation Results

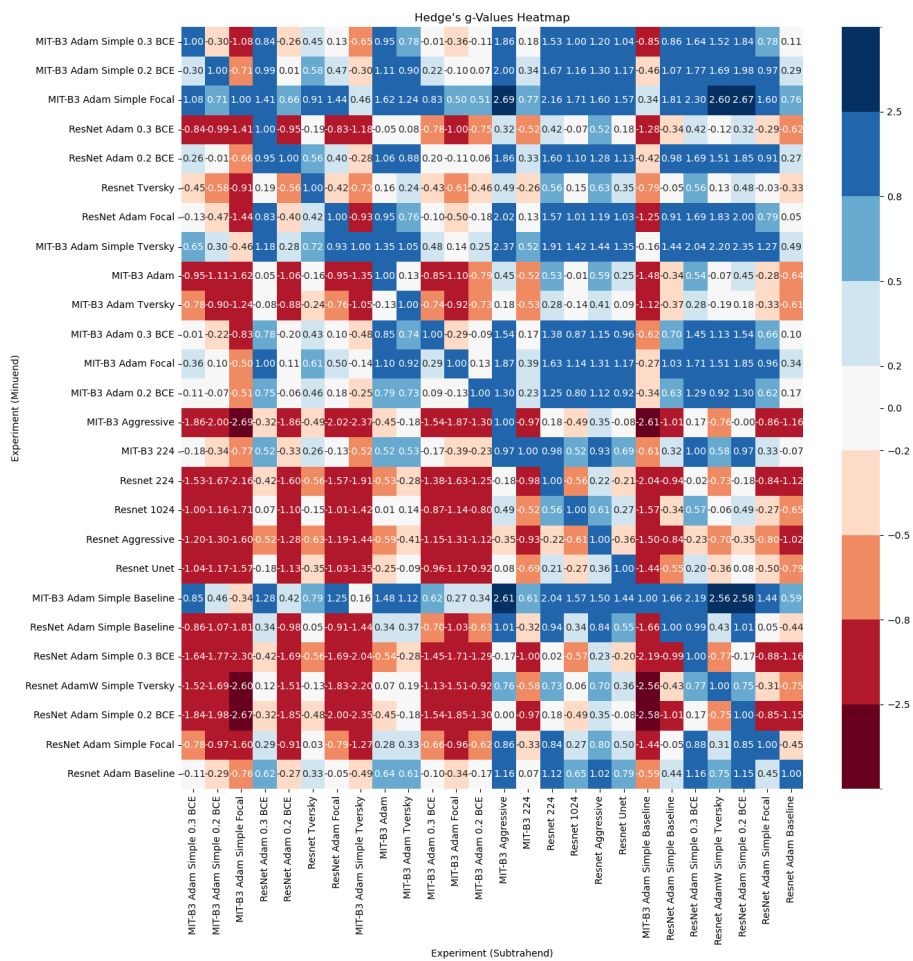


Figure D.1: Dice Score Hedge's g-value Heatmap

D Segmentation Results

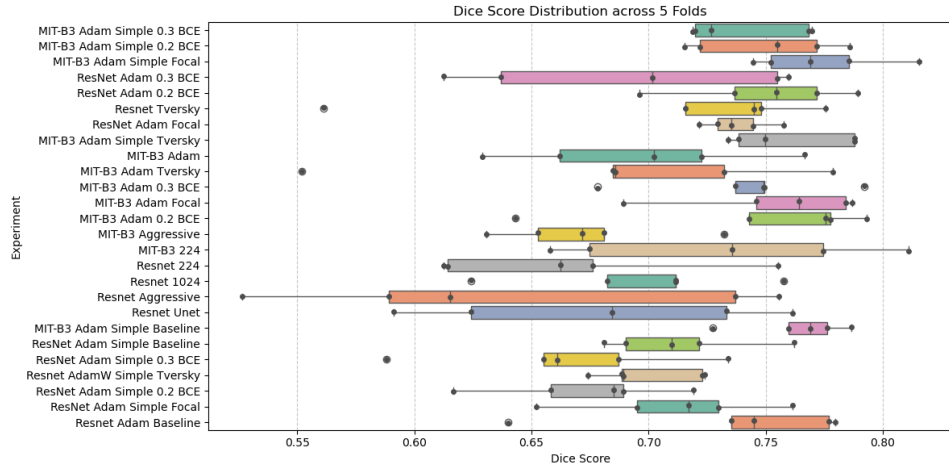


Figure D.2: Dice Score Boxplot. Certain models demonstrate high spread while others are more stable.

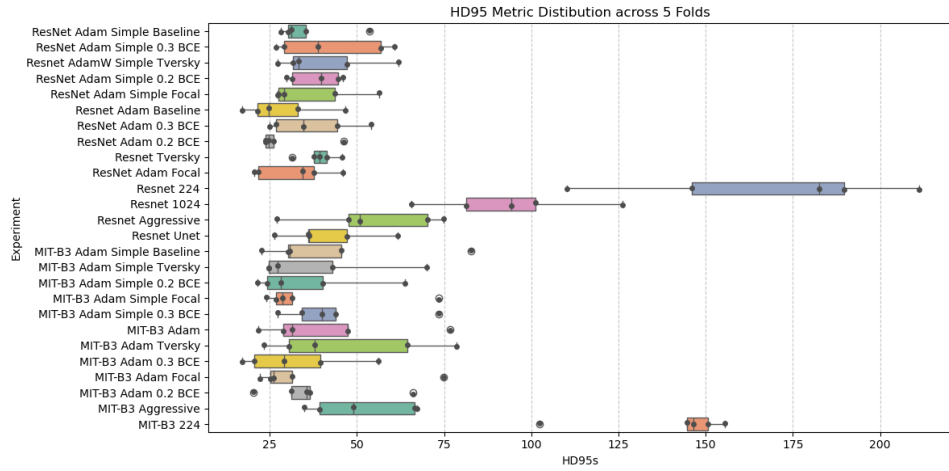


Figure D.3: Hedge's g-value Boxplot

D Segmentation Results

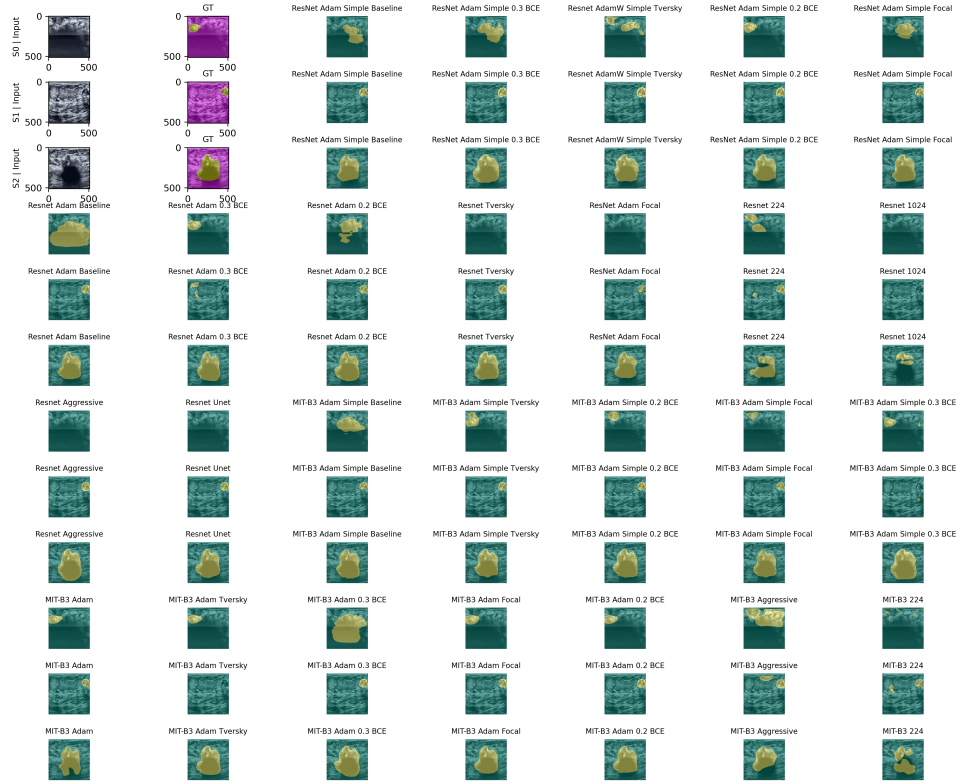


Figure D.4: Sample Segmentation Visualizations on all Models

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